

# AI Technology and Society

2024 Edition

Generative AI | Industry Impact | Ethics | Future Outlook

# Table of Contents

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|                                                    |   |
|----------------------------------------------------|---|
| <a href="#">Chapter 1: Generative AI Trends</a>    | 3 |
| <a href="#">Chapter 2: Industry Impact</a>         | 4 |
| <a href="#">Chapter 3: Ethics &amp; Regulation</a> | 5 |
| <a href="#">Chapter 4: Future Outlook</a>          | 6 |
| <a href="#">References &amp; External Links</a>    | 7 |

## Chapter 1: Generative AI Trends

The release of ChatGPT in late 2022 triggered rapid mainstream adoption of generative AI across industries worldwide.

Key Generative AI Categories:

- Large Language Models (LLMs): Text generation, summarization
- Image Generation: Stable Diffusion, DALL-E, Midjourney
- Multimodal AI: Cross-modal text, image, and audio processing

The pace of development far exceeds previous technological revolutions.

## Chapter 2: Industry Impact

AI is driving transformative change across all major industries.

Sector-by-Sector Impact:

- Manufacturing : Predictive maintenance, quality inspection AI
- Finance : Fraud detection, investment analysis
- Healthcare : Medical imaging, drug discovery acceleration
- Retail : Demand forecasting, personalized recommendations
- Legal : Contract review, case law research

## Chapter 3: Ethics & Regulation

Rapid AI adoption has surfaced significant ethical and regulatory questions.

Key Regulatory Developments:

- EU AI Act (2024): Risk-based AI regulation framework
- US Executive Order (2023): AI safety and security standards
- G7 Hiroshima AI Process: International voluntary code of conduct

Organizations should proactively develop internal AI ethics guidelines.

## Chapter 4: Future Outlook

AI will continue to reshape society in profound ways through 2030 and beyond.

Key Trends to Watch (2025-2030):

- Progress toward Artificial General Intelligence (AGI)
- Edge AI: Real-time on-device processing at scale
- International regulatory harmonization
- Established human-AI collaboration models

Organizations that adapt strategically will capture competitive advantage.

References & External Links

Tap a link to open in the browser:

- OpenAI Research Blog

<https://openai.com/research>

- Google DeepMind Publications

<https://deepmind.google/research/>

- EU Artificial Intelligence Act

<https://artificialintelligenceact.eu/>

- Stanford AI Index Report

<https://aiindex.stanford.edu/>

- MIT Technology Review - AI

<https://www.technologyreview.com/topic/artificial-intelligence/>